

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 1.1: Taxonomy I — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 1.1: Taxonomy I
Driving Question	DRIVING QUESTION: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Introduction to Taxonomy and Classification: Why Do Our Groupings Differ?	We sorted eight familiar organisms (goat, chicken, maize plant, mushroom, algae, bacteria, bean plant, mosquito) using our own rules. Different groups made different groupings and placed the same organism in different groups. The same organism also had different names in Kikuyu, Luo, Kiswahili and English.	Organisms can be grouped in many ways depending on the criteria chosen. Taxonomy is the science of naming, describing and classifying organisms. Local names and personal criteria are useful but cannot be used by everyone in the same way.	Each group was correct by its own criteria, but because criteria and names differ, people cannot be sure they are discussing the same organism. Scientists therefore need a universal, agreed system of classification so that everyone can identify, study and communicate about organisms in the same way.
Lesson 2 Importance of Classification: Why Do We Need to Identify Organisms Correctly?	What did we observe in the cases of edible and poisonous mushrooms, useful and harmful insects, and crops and weeds? Which features helped us identify each organism, and what happened when identification was wrong?	What did we learn about why classification is important for identifying, studying and communicating about organisms, and for conserving biodiversity? Why are local names alone not enough?	How can we explain, using Claim-Evidence-Reasoning, why scientists need a shared system of classification? What new question does this raise about common levels or groups?
Lesson 3 Taxonomic Units: From Kingdom to Species	We used organism cards and category cards to arrange the ranks from broadest to most specific: Kingdom, Phylum/Division, Class, Order, Family, Genus and Species. We watched the video on classification of living organisms and saw how the lion (<i>Panthera leo</i>) and maize (<i>Zea mays</i>) fit into the ranks. As we moved down, there were fewer organisms in each group and they shared more features.	The taxonomic ranks are nested: each rank sits inside the one above it. Kingdom is the broadest group and Species is the most specific. Every organism can be given one exact place in this hierarchy, no matter what name it has in Kikuyu, Luo, Kiswahili or English.	This helps solve our puzzle because a ranked, universal hierarchy lets everyone in Kenya and the world place, identify and talk about the same organism in the same way, unlike sorting by habitat, colour or usefulness. We still need to learn how the ranks relate in more detail from the largest group to the smallest and how organisms are named with Genus and species.

Lesson 4 Hierarchy of Classification: Climbing the Taxonomic Staircase	We observed that organisms can be arranged into seven ranked groups (Kingdom, Phylum, Class, Order, Family, Genus, Species). When we classified the goat, chicken, maize plant, bean plant, mosquito and mushroom, some shared the same Kingdom or Phylum but separated at lower levels.	We learned that taxonomic groups form a hierarchy from broad to specific: each step down has fewer organisms that are more similar to each other. Species is the most specific level. Using a common chart or reference lets people classify organisms the same way.	This helps explain the opening puzzle: sorting by colour, habitat or local names (Kikuyu, Luo, Kiswahili, English) gives different groups, but a shared ranked hierarchy lets everyone place an organism in the same position. We still need a way to give each species a unique name recognised worldwide: How can scientists give one particular organism a unique name that is recognised worldwide?
Lesson 5 Scientific Names and Binomial Nomenclature: One Name for Every Organism	We compared the common names of organisms such as maize, beans, goat and mosquito in Kikuyu, Luo, Kiswahili and English and found they differ between communities. We watched a video and read about the two-part naming system and its rules.	Scientific names give every organism one universal name that scientists everywhere can use. A scientific name has two parts, the genus (capital letter) and the species (small letters), written in Latin form and in italics or underlined separately, e.g. <i>Zea mays</i> .	Common names vary with language and community, which causes confusion when sharing information, so a universal binomial name is needed. Because every organism has just one agreed scientific name, people everywhere can identify, study and communicate about it in the same way, which helps solve the puzzle from our original sorting activity.
Lesson 6 Applying Binomial Nomenclature: Naming Our Local Organisms the Same Way Everywhere	We observed that the same organism (such as maize, goat or mosquito) has different local names but only one two-part scientific name (for example <i>Zea mays</i> , <i>Capra hircus</i> , <i>Anopheles gambiae</i>), written with a capital genus, a lower-case species, and italics or underlining.	We learned the principles of binomial nomenclature: two names (genus and species), correct capitalisation, italics or underlining, and the use of Latin so the name is the same worldwide.	We can explain that scientific names give a universal way of identifying organisms, so people who use different local names (Kikuyu, Luo, Kiswahili, English) can still be sure they are talking about the same organism, which solves part of our driving question.
Lesson 7 Five Kingdoms of Living Organisms	What did you observe about the cells, organisation and feeding of organisms in each of the five kingdoms (for example bacteria in maziwa lala, <i>Spirogyra</i> , mushroom, maize and bean plants, goat, chicken and mosquito)?	What characteristics do scientists use to group organisms into the five kingdoms, and how are they different from colour, habitat or usefulness?	How does the five-kingdom system solve the problem of organisms having different groupings and names in Kikuyu, Luo, Kiswahili and English, and what question does it raise about classifying unfamiliar organisms?
Lesson 8 Classification into the Five Kingdoms and Importance of Taxonomy: Our Final Explanation	We observed that unfamiliar organisms such as <i>Amoeba</i> , <i>Spirogyra</i> , <i>Rhizopus</i> , a mushroom, bacteria, maize, a tilapia and an <i>Anopheles</i> mosquito could be placed into one of five kingdoms (Monera, Protocista, Fungi, Plantae, Animalia) using shared observable evidence: cell type, number of cells, cell wall, chlorophyll, and mode of nutrition. We also observed that our Lesson 1 groupings by habitat, colour, size or usefulness placed the same organisms in different groups and local names differed.	We learned that classification groups organisms using shared characteristics in a hierarchy (kingdom, phylum, class, order, family, genus, species); that binomial nomenclature gives every species a unique two-part scientific name (genus capitalised, species lowercase) understood worldwide; and that taxonomy is important for identifying disease vectors, crop pests and useful organisms, understanding the economic importance of Monera, Protocista and Fungi, and conserving	Living organisms can be systematically organised into hierarchical groups from five kingdoms down to species, named using a universal two-part scientific name, and identified through shared observable characteristics. Because everyone uses the same evidence-based criteria and names, people in Nairobi, Kisumu, Mombasa and across the world can identify, study and communicate about the same organism in the same way, which resolves the puzzle from Lesson 1.

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